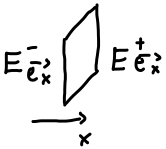
Distribuce v $\mathbb{R}^1$

→ chceme popsat singulární zdroje

- bodové, křivkové, plošné

→ chceme popsat nespojitost polí

- např.:



$$\operatorname{div} \vec{E} = \frac{dE_x}{x}$$

→ kvantová mechanika → vlastní stavy operátoru polohy $\hat{x}$

Problém: hustota bodového náboje ρ v $x = 0$

$$\Rightarrow \quad \rho = \begin{cases} 0 & x \neq 0 \\ \text{něco} & x = 0 \end{cases}$$

a zároveň $\int_{\mathbb{R}} \rho \, dx = q$

→ problém s Lebesgueovským integrálem, pokud ρ je funkce

$\Rightarrow \int_{\mathbb{R}} \delta \, dx = 1$ → funkcionální definice

-> Schwartz 1950 - Fields medal

Spojitý lineární funkcionál na vhodném prostoru testovacích funkcí

$\mathcal{D}$ ... prostor testovacích funkcí

$\mathcal{D}'$ ... prostor distribucí

$$\phi \in \mathcal{D},\ T \in \mathcal{D}' \Rightarrow T : \mathcal{D} \to \mathbb{R}$$

$$T(\phi) = \langle T, \phi \rangle \in \mathbb{R}$$

- spojitost $\phi_n \xrightarrow{\mathcal{D}} \phi \Rightarrow T(\phi_n) \xrightarrow{\mathbb{R}} T(\phi)$
- linearita $T(\alpha \phi_1 + \beta \phi_2) = \alpha T(\phi_1) + \beta T(\phi_2)$

$$T(\phi) = T_x \phi^x = \int_{\mathbb{R}} T_x \phi^x \, dx$$

$$\phi^x = \phi(x)$$

Prostory testovacích funkcí

Hladké funkce s kompaktním nosičem

$$\mathcal{D} = \{ \phi \in C^\infty \mid \operatorname{supp} \phi \text{ kompaktní} \}$$

$\phi_n \xrightarrow{\mathcal{D}} \phi \Rightarrow \exists K$ (kompakt) : $\forall n$

$\operatorname{supp} \phi_n \subset K$: $\phi_n^{[k]} \overset{K}{\rightrightarrows} \phi^{[k]} \quad \forall k \in \mathbb{N} \cup \{0\}$

$\mathcal{D}'$ ... distribuce s neomezeným růstem

Hladké funkce rychle klesající

$\mathcal{S} = \{ \phi \in C^\infty \mid \forall m \quad |x|^m \phi$ je omezená $\}$

$\phi_n \xrightarrow{\mathcal{S}} \phi \Rightarrow \forall m, k \quad |x|^m \phi_n^{[k]} \overset{\mathbb{R}}{\rightrightarrows} |x|^m \phi^{[k]}$

$\mathcal{S}'$ ... Schwarzovské (temperované) distribuce

→ pomalu rostoucí

Hladké funkce

$\mathcal{E} := \{ \phi \in C^\infty(\mathbb{R}) \}$

$\phi_n \xrightarrow{\mathcal{E}} \phi \Rightarrow \forall K$ (kompakt) $\forall k \quad \phi_n^{[k]} \overset{K}{\rightrightarrows} \phi^{[k]}$

$\mathcal{E}'$ ... distribuce s kompaktním nosičem

Vztahy:

$\mathcal{D} \subset \mathcal{S} \subset \mathcal{E} \qquad \mathcal{E}' \subset \mathcal{S}' \subset \mathcal{D}'$

$L^2 = L^{2'}$ hochu, to ti řekne Rieszova věta

Vlastnosti distribucí

Rovnost

$$T_1 \overset{\mathcal{D}'}{=} T_2 \quad \Leftrightarrow \quad \langle T_1, \phi \rangle = \langle T_2, \phi \rangle \quad \forall \phi \in \mathcal{D}$$

→ rovnost na množině $\Omega \subset \mathbb{R}$

$$T_1 \overset{\Omega}{=} T_2 \quad \Leftrightarrow \quad \langle T_1, \phi \rangle = \langle T_2, \phi \rangle \quad \forall \phi \in \mathcal{D} : \operatorname{supp} \phi \subset \Omega$$

Nosič distribuce

$$\operatorname{supp} T = \mathbb{R} \setminus \bigcup_{\Omega :\, T \overset{\Omega}{=} 0} \Omega$$

identicky nulová distribuce

Konvergence

$$T_n \xrightarrow{\mathcal{D}'} T \quad \Leftrightarrow \quad \langle T_n, \phi \rangle \xrightarrow{\mathbb{R}} \langle T, \phi \rangle \quad \forall \phi \in \mathcal{D}$$

Příklady distribucí

Diracova δ funkce (distribuce)

$$\langle \delta, \phi \rangle = \phi(0)$$

ve fyzice : $\int_{\mathbb{R}} \delta(x)\, \phi(x)\, dx = \phi(0)$

Posunutá δ funkce

$$\langle \delta_a, \phi \rangle = \langle \delta(x-a), \phi(x) \rangle = \phi(a)$$

Regulární distribuce

- generované $L^1_{loc}(\mathbb{R})$ - lokálně integrabilní funkce na $\mathbb{R}$

$f \in L^1_{loc}(\mathbb{R})$

$$\langle T_f, \phi \rangle \text{"="} \langle f, \phi \rangle = \int_{\mathbb{R}} f(x)\,\phi(x)\,dx$$

$$= \int_{\operatorname{supp}\phi} f(x)\,\phi(x)\,dx$$

je kompakt $\{ \operatorname{supp}\phi$

=) existuje vždy

Heavisideova distribuce

$$\langle T_\theta, \phi \rangle = \langle \theta, \phi \rangle = \int_0^\infty \phi(x)\,dx$$

Derivace distribuce

Motivace

$$T_f' = T_{f'}$$

supp ϕ je kompakt

$$\langle T_{f'}, \phi \rangle = \int_{\mathbb{R}} f' \phi \, dx = \underbrace{\left[f \phi \right]_{-\infty}^{\infty}}_{0} - \int_{\mathbb{R}} f \phi' \, dx$$

$$= - \langle T_f , \phi' \rangle$$

Definice

$$\langle T', \phi \rangle = - \langle T, \phi' \rangle$$

$$\langle T'', \phi \rangle = \langle T, \phi'' \rangle$$

$$\vdots$$

$$\langle T^{(n)}, \phi \rangle = (-1)^n \langle T, \phi^{(n)} \rangle$$

Př

$$\Theta' \, ? \quad \Rightarrow \quad \langle \Theta', \phi \rangle = - \langle \Theta, \phi' \rangle$$

$$= - \int_0^{\infty} \phi'(x) \, dx$$

$$= - \left[\phi \right]_0^{\infty} dx = \phi(0)$$

$$= \langle \delta, \phi \rangle$$

$$\Rightarrow \quad \Theta' = \delta$$

Př - Fyz

$$\operatorname{div} \vec{E} = \frac{\varrho}{\varepsilon_0} \xrightarrow{1D} \frac{\partial E}{\partial x} = \frac{\varrho}{\varepsilon_0}$$

Př

$$\langle \delta^{(n)}, \phi \rangle = (-1)^n \langle \delta, \phi^{(n)} \rangle = (-1)^n \phi^{(n)}(0)$$

Násobení distribucí hladkou funkcí

$T \in \mathcal{D}' \quad f \in C^\infty$

$$\langle fT, \phi \rangle = \langle T, f\phi \rangle$$

Př

$$\langle f\delta, \phi \rangle = \langle \delta, f\phi \rangle = f(0)\phi(0) = f(0) \langle \delta, \phi \rangle$$

$$\langle f\delta', \phi \rangle = \langle \delta', f\phi \rangle = -\langle \delta, f'\phi + f\phi' \rangle =$$

$$= -f'(0)\phi(0) - f(0)\phi'(0) =$$

$$= -f'(0) \langle \delta, \phi \rangle + f(0) \langle \delta', \phi \rangle$$

$$f\delta' \overset{\mathcal{D}}{=} f(0)\delta - f'(0)\delta$$

Př - Derivace nehladké funkce v $x = 0$

$$f(x) = f_+(x)\,\Theta(x) + f_-(x)\,\Theta(-x)$$

$$\langle (f_+(x)\Theta(x))' , \phi \rangle = -\langle f_+(x)\Theta(x) , \phi' \rangle$$

$$= -\langle \Theta(x) , f_+(x)\phi' \rangle = -\int_0^\infty f_+(x)\,\phi'(x)\,dx$$

$$= -\Big[f_+(x)\,\phi(x) \Big]_0^\infty + \int_0^\infty f_+'(x)\,\phi(x)\,dx$$

$$= \langle f_+(0)\delta , \phi \rangle + \langle f_+'\Theta , \phi \rangle$$

$$(f_+\Theta)' \overset{\mathcal{D}'}{=} f_+(0)\,\delta + f_+'\,\Theta$$

$$(f_-\Theta(-x))' \overset{\mathcal{D}'}{=} -f_-(0)\,\delta + f_-'\,\Theta(-x)$$

$$[f] = f_+(0) - f_-(0)$$

$$f^{[1]} = f_+'\,\Theta(x) + f_-'\,\Theta(-x)$$ regularized dist derivative

$$f' = [f]\,\delta + f^{[1]}$$

$$f'' = [f]\,\delta' + [f']\,\delta + f^{[11]}$$

$$xT \overset{\mathcal{D}'}{=} 0$$

$$\Updownarrow$$

$$T = c\delta \qquad c \in \mathbb{R}$$

Dk "$\Downarrow$" nechť $\omega \in \mathcal{D} : \omega(0) = 1$ je pevná

$$\forall \phi \in \mathcal{D} \quad \exists \psi \in \mathcal{D} : \quad \phi = \phi(0)\omega + x\psi$$

$$\psi(0) = \lim_{x \to 0} \frac{\phi(x) - \phi(0)\,\omega(x)}{x}$$

$$\langle T, \phi \rangle = \phi(0) \langle T, \omega \rangle + \langle T, x\psi \rangle$$

$$= c\,\phi(0) + \underbrace{\langle xT, \psi \rangle}_{\overset{\|}{0}} = c \langle \delta, \phi \rangle$$

$$\Rightarrow \quad T \overset{\mathcal{D}'}{=} c\delta$$

Př - distribuce $\ln|x|$

Platí $\ln|x| \in L^1_{loc}(\mathbb{R})$?

$$\lim_{\varepsilon \to 0} \int_\varepsilon^1 \ln|x| \, dx = \lim_{\varepsilon \to 0} \left[x \ln x - x \right]_\varepsilon^1$$

$$= \lim_{\varepsilon \to 0} \left(-\varepsilon \ln \varepsilon - 1 + \varepsilon \right) = -1$$

$$\Rightarrow \qquad \ln|x| \in L^1_{loc}(\mathbb{R})$$

$$\langle \ln|x|, \phi \rangle = \int_{\mathbb{R}} \ln|x|\, \phi(x)\, dx$$

Př. - distribuce v.p. $\frac{1}{x}$

v.p. ... valeur principal – hlavní hodnota

$$\left\langle \text{v.p.}\, \frac{1}{x}, \phi \right\rangle = \lim_{\varepsilon \to 0} \int_{\mathbb{R} \setminus (-\varepsilon, \varepsilon)} \frac{\phi(x)}{x}\, dx$$

$$= \lim_{\varepsilon \to 0} \int_{\varepsilon}^{\infty} \frac{\phi(x) - \phi(-x)}{x}\, dx$$

$$\langle (\ln|x|)', \phi \rangle = - \langle \ln|x|, \phi' \rangle =$$

$$= - \int_{\mathbb{R}} \ln|x|\, \phi'(x)\, dx = - \lim_{\varepsilon \to 0} \left(\int_{-\infty}^{-\varepsilon} \dots + \int_{\varepsilon}^{\infty} \dots \right)$$

$$\lim_{\varepsilon \to 0} \int_{\varepsilon}^{\infty} \ln|x|\, \phi(x)\, dx = \lim_{\varepsilon \to 0} \left(\left[\ln|x|\, \phi(x) \right]_{\varepsilon}^{\infty} - \int_{\varepsilon}^{\infty} \frac{\phi(x)}{x}\, dx \right)$$

$$= \lim_{\varepsilon \to 0} \left(- \ln|\varepsilon|\, \phi(\varepsilon) - \int_{\varepsilon}^{\infty} \frac{\phi(x)}{x}\, dx \right)$$

$$\lim_{\varepsilon \to 0} \int_{-\infty}^{-\varepsilon} \ln|x|\, \phi'(x)\, dx = \lim_{\varepsilon \to 0} \left(\ln|\varepsilon|\, \phi(-\varepsilon) - \int_{\infty}^{-\varepsilon} \frac{\phi(x)}{x}\, dx \right)$$

$$\underbrace{-\ln|\varepsilon|\,\varepsilon}_{\Downarrow\ 0}\ \frac{\phi(\varepsilon)-\phi(-\varepsilon)}{\varepsilon}$$

$$\Rightarrow \qquad (\ln|x|)' \overset{\mathcal{D}'}{=} \text{v.p.}\,\frac{1}{x}$$

Substituce v distribuci

Motivace skrze regulární distribuce

$$T_f \longrightarrow T_{f\circ g}\,,\ \text{tj.}\quad T_{f(y)} \to T_{f(g(x))}$$

$$\langle T_{f(g(x))}, \phi(x)\rangle = \int_{\mathbb{R}} f(g(x))\,\phi(x)\,dx$$

$$\left[\, g(x) = y \quad \Rightarrow \quad dx = \frac{1}{g'(g^{-1}(y))}\,dy \,\right]$$

g je prostá!

$$= \int_{\mathbb{R}} f(y)\,\phi(g^{-1}(y))\,\frac{dy}{|g'(g^{-1}(y))|}$$

$\Rightarrow$ pro $T \in \mathcal{D}'$

$$\langle T\circ g, \phi\rangle = \langle T(f), \phi\rangle = \left\langle T, \frac{\phi\circ g^{-1}}{|g'\circ g^{-1}|}\right\rangle$$

Př

$$\langle \delta\circ g, \phi\rangle = \langle \delta(g), \phi\rangle = \left\langle \delta, \frac{\phi\circ g^{-1}}{|g'\circ g^{-1}|}\right\rangle$$

$$= \frac{\phi\circ g^{-1}(0)}{|g'\circ g^{-1}(0)|} = \sum_{x_i : g(x_i)=0} \frac{\phi(x_i)}{|g'(x_i)|}$$

$$= \sum_{x_i : g(x_i)=0} \frac{1}{|g'(x_i)|} \langle \delta_{x_i}, \phi\rangle$$

$$\delta\circ g = \delta(g(x)) = \sum_{x_i} \frac{1}{|g'(x_i)|} \underset{\substack{\uparrow \\ \delta(x-x_i)}}{\delta_{x_i}}$$

Greenovy funkce v 1D

→ řešení pro bodový zdroj

⇒ superp. → řešení pro lib. rozložení

$$L_x f(x) = g(x)$$

$$L_x G = \delta$$

$$L_x G(x,x') \overset{\partial!}{=} \delta(x,x')$$

$$f(x) = \langle G(x,x'), g(x')\rangle$$

$$L_x f(x) = \left\langle L_x G(x,x'), g(x') \right\rangle = \left\langle \delta(x-x'), g(x') \right\rangle = g(x)$$

$$f(x) = \int G(x,x')\, g(x')\, dx'$$

$$L_x f = \int L_x G(x,x')\, g(x')\, dx' = g(x)$$

- pozn.

jádro

$$L_x \overset{\downarrow}{h}(x) = 0$$

$$G(x,x') \longrightarrow \tilde{G}(x,x') = G(x,x') + h(x)$$

→ nejednoznačnosti se zbavíme o. p.

- porovnání

$$L_x = \frac{d}{dx} \quad \Rightarrow \quad G(x,x') = \Theta(x-x')$$

$$L_x \Theta(x-x') = \delta(x-x')$$

Problém:

L_x je lineární samosdružený dif. operátor

$(Af, g) = (f, A^\dagger g)$ $\quad A = A^\dagger$

$I \subset \mathbb{R}$ ... interval řešení

$$\frac{d}{dx} \to i\,\frac{d}{dx}$$

$B_{\partial I}$ ... o. p

$B_{\partial I} f = C \Rightarrow a$

- $f|_{\partial I} = 0$ … Dirichletovy
- $f'|_{\partial I} = 0$ … Neumannovy

Věta

Nechť $L_x = \sum_{n=0}^{k} a_n(x) \frac{d^n}{dx^n}$ lin. samosdružený dif. op. st. k, $I \subset \mathbb{R}$, $B_{\partial I}$.

Pak nechť

1) $\exists\, g_y^{\pm}(x)$ takové, že:

$$L_x g_y^+(x) = 0 \qquad x \geq y$$

$$L_x g_y^-(x) = 0 \qquad x \leq y$$

2) $B_{\partial I^+} g^+ = 0 \qquad B_{\partial I^-} g^- = 0 \qquad \partial I^{\pm}$ pravá/levá hranice

3) $\forall n \in \{0, \dots, k-2\} \qquad g_y^{+(n)}(y) = g_y^{-(n)}(y)$

4) $g_y^{+(k-1)}(y) - g_y^{-(k-1)}(y) = \frac{1}{a_k(y)}$

$$\Rightarrow \quad G(x,y) = g_y^+(x)\,\Theta(x-y) + g_y^-(x)\,\Theta(y-x)$$

Dh

$$\frac{d}{dx} G(x,y) = g'^{+}_y(x)\,\Theta(x-y) + g'^{-}_y(x)\,\Theta(y-x)$$

$$+ \; g_y^+(x)\,\delta(x-y) + g_y^-(x)\,\delta(y-x)$$

$$= g_y^+(x)\,\delta(x-y) - g_y^-(x)\,\delta(x-y)$$

$$\frac{d^n}{dx^n} G(x,y) = g_y^{+(n)}(x)\,\Theta(x-y) + g_y^{-(n)}(x)\,\Theta(y-x)$$

$$+ \text{"}\, g^{(n-1)}\delta + \cancel{g^{(n-2)}\delta'} + \dots$$

$$+ \; \cancel{g\,\delta^{(n-1)}} \,\text{"}$$

$$\downarrow$$

$$\left(g_y^{+(n-1)}(x) - g_y^{-(n-1)}(x) \right) \delta(x-y)$$

$$= \frac{1}{a_n(y)}\,\delta(x-y)$$

$$L_x f(x) = g(x) \qquad f(x) = \int G(x,y)\,g(y)\,dy$$

$L_x = -\frac{d^2}{dx^2} + h^2$ $\qquad I = (0,d)$ $\qquad B_{\partial I} f = f|_{\partial I} = 0$

$L_x g^{\pm} = 0 \quad \Rightarrow \quad$ FS: e^{-hx}, e^{hx}

$g_y^{\pm}(x) = A^{\pm}(y)\, e^{hx} + B^{\pm}(y)\, e^{-hx}$

$B_{\partial I}$: $\qquad A^{+}(y)\, e^{hd} + B^{+}(y)\, e^{-hd} = 0$

$$\Rightarrow \quad B^{+}(y) = \overset{?}{-}\,\overset{\checkmark}{A}{}^{+}(y)\, e^{2hd}$$

$$A^{-}(y) = -B^{-}(y)$$

spoj $g^{\pm}$:

$$A^{+}(y)\left(e^{hy} - e^{2hy} e^{-hy}\right) = A^{-}(y)\left(e^{hy} - e^{-hy}\right)$$

skok $g^{\pm\prime}$:

$$h A^{+}(y)\left(e^{hy} + e^{2hy} e^{-hy}\right)$$
$$- h A^{-}(y)\left(e^{hy} + e^{-hy}\right) = \frac{1}{a_2(y)} = -1$$

$$\Rightarrow \quad A^{+}(y) = -\frac{e^{-hd}}{2h}\,\frac{\sinh(hy)}{\sinh(hd)}$$

$$A^-(y) = \frac{1}{2h}\,\frac{\sinh(h(y-d))}{\sinh(hd)}$$

$$G(x,y) = -\frac{1}{h}\frac{\sinh(hy)}{\sinh(hd)}\,\sinh(h(x-d))\;\theta(x-y)$$
$$-\frac{1}{h}\,\frac{\sinh[h(y-d)]}{\sinh(hd)}\,\sinh(hx)\;\theta(y-x)$$

$$G_0(x,y) = \lim_{h\to 0} G(x,y) = -\frac{yx}{d} + y\,\theta(x-y)$$
$$+\,x\,\theta(y-x)$$

často se setkáme s $L_x = -\frac{d^2}{dx^2} + \frac{1}{x}$ na $(0,d)$

$\Rightarrow$ vyřešíme na (ε,d) a na hraniční podmínce limita $\varepsilon \to 0^+$ a ověříme [illegible].

L_x … samosdružený lineární diferenciální operátor

$L_x \psi_n = \lambda_n \psi_n$, $I \subset \mathbb{R}$, $B_{\partial I}$

Relace ortogonality:

$$\int_I \psi_n(x)\overline{\psi_m}(x)\,dx = \delta_{nm}$$

Relace úplnosti

$$\sum_{n=0}^{\infty} \psi_n(x)\overline{\psi_n}(x') = \delta(x-x')$$

$$\to \int \sum_{n=0}^{\infty} \psi_n(x)\,\underbrace{\overline{\psi_n}(x)\, f(x)\, dx}_{c_n} = \int \delta(x-x')\, f(x)\, dx$$

$$= f(x')\, dx$$

$\Rightarrow$ projektor $\sum |\psi_n\rangle\langle\psi_n|$

$\Rightarrow$ Greenova funkce

$$G(x,y) = \sum_{n=0}^{\infty} c_n(y)\,\psi_n(x)$$

$$L_x G(x,y) = \sum_{n=0}^{\infty} c_n(y)\,\lambda_n \psi_n(x) \overset{?}{=} \delta(x-y) \quad \Big|\,\overline{\psi_m}(x)\,\Big| \int_I dx$$

$$\downarrow$$

$$\int \sum_{n=0}^{\infty} c_n(y)\,\lambda_n\, \overline{\psi_m}\psi_n \, dx = \int \overline{\psi_m}(x)\,\delta(x-y)\, dx$$

$$\sum_{n=0}^{\infty} c_n(y)\,\lambda_n \delta_{mn} = \overline{\psi_m}(y)$$

$$\lambda_m c_m(y) = \overline{\psi_m}(y)$$

$$\Rightarrow \quad G(x,y) = \sum_{n=0}^{\infty} \frac{\psi_n(x)\,\overline{\psi_n}(y)}{\underset{\uparrow\, = 0}{\lambda_n}}$$

Co když existuje homo řešení ?

$$L_x = -\frac{d^2}{dx^2} \qquad I = S^1 \qquad x \in (0,1)$$

$$\psi(0) = \psi(1)$$

$L_x c = 0$ → jakákoli konstanta řeší tent. problém

$\Rightarrow \lambda_0 = 0$

$\Rightarrow$ zahodím nultý člen:

$$\tilde{G}(x,y) = \sum_{n=1}^{\infty} \frac{\psi_n(x)\overline{\psi_n}(x)}{\lambda_n}$$

↓ nutno zúplnit

$$\Rightarrow \quad L_x \tilde{G}(x,y) = \delta(x-y) - \underbrace{\psi_0(x)\overline{\psi_0}(y)}_{\text{nějaká! konstanta}}$$